Power Electronics

DC-AC Converters (Inverters)

逆变器

Dr Atif Jafri
Atif.Jafri@glasgow.ac.uk

Please read Chapter 8 in the textbook

Schemes of Inverters to study

Phase Output	Bridge	Switching Schemes	
Single Phase	Half	Square Wave	• Operation in different switching schemes • Output Waveform • Harmonics • Practical Circuit • Avoiding shoot through
		Pules Width Modulation	
	Full	Square Wave	
		Pules Width Modulation (Bipolar & Unipolar)	
Three Phase	3-Leg	Square Wave	
		Pules Width Modulation	

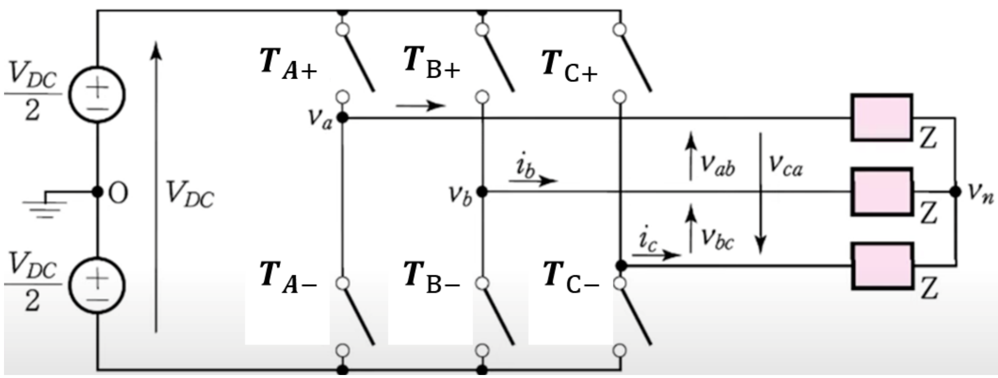
So far studied Single Phase Inverter

Schemes of Inverters to study

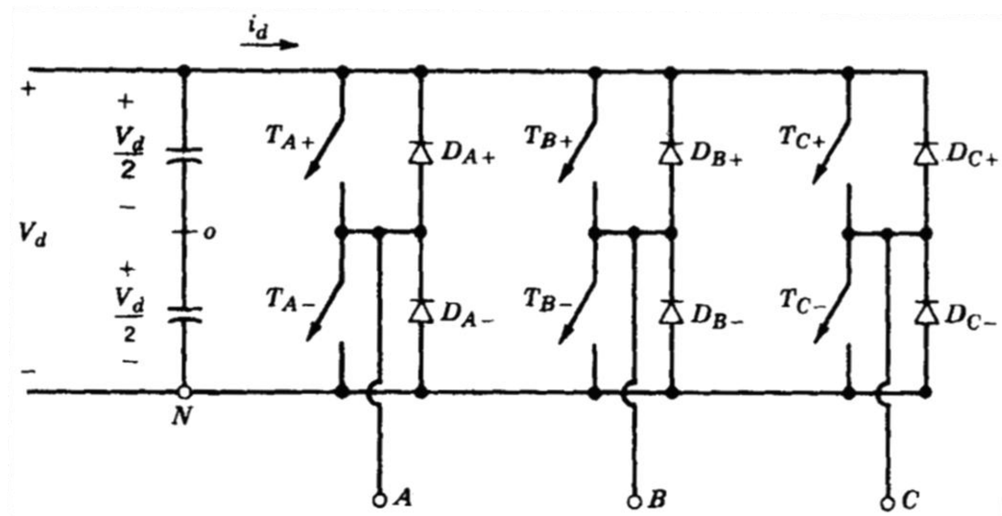
<table>
<tr><th>Phase Output</th><th>Bridge</th><th colspan="2">Switching Schemes</th><td rowspan="8">Ideal Circuit Diagram
Operation
Output Waveform
Harmonics
Practical Circuit Diagram
Avoiding shoot through</td></tr>
<tr><td rowspan="5">Single Phase</td><td rowspan="2">Half</td><td colspan="2">Square Wave</td></tr>
<tr><td colspan="2">Pules Width Modulation</td></tr>
<tr><td rowspan="3">Full</td><td colspan="2">Square Wave</td></tr>
<tr><td colspan="2">Pules Width Modulation</td></tr>
<tr><td>Bipolar</td><td>Unipolar</td></tr>
<tr><td rowspan="2">Three Phase</td><td rowspan="2">3-Leg</td><td colspan="2">Square Wave</td></tr>
<tr><td colspan="2">Pules Width Modulation</td></tr>
</table>

Inverter Types

- 3-Leg for 3 Phase Inveter



3 Phase Inverter (Ideal)

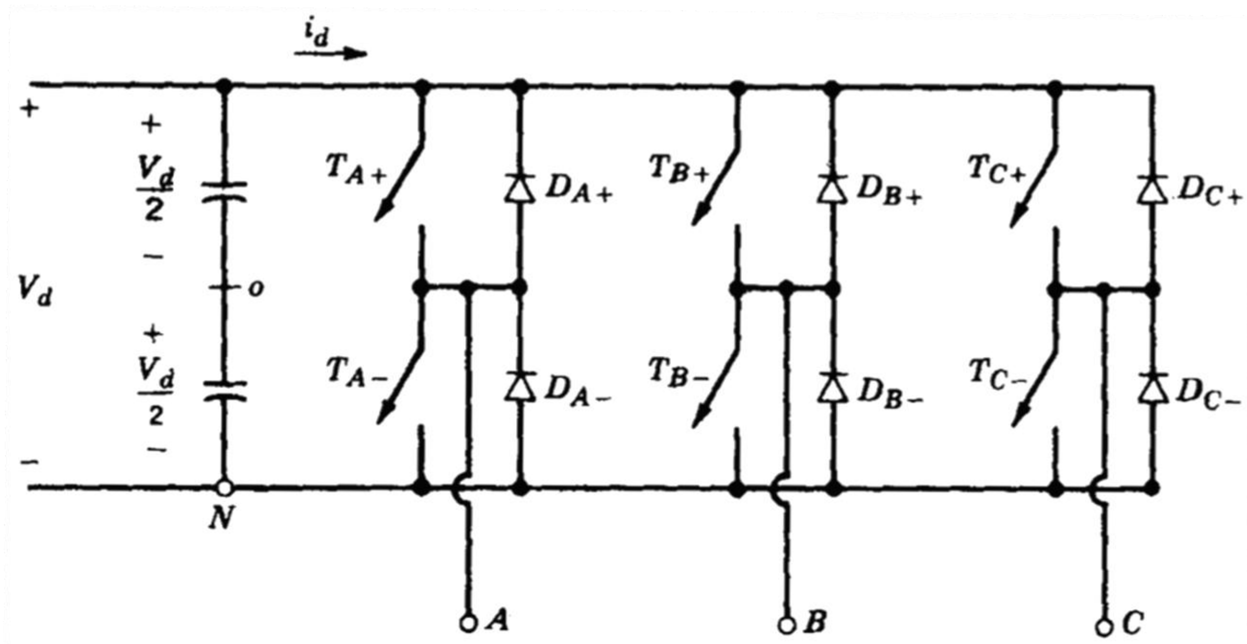


3-Phase Inverter with Flyback Diodes

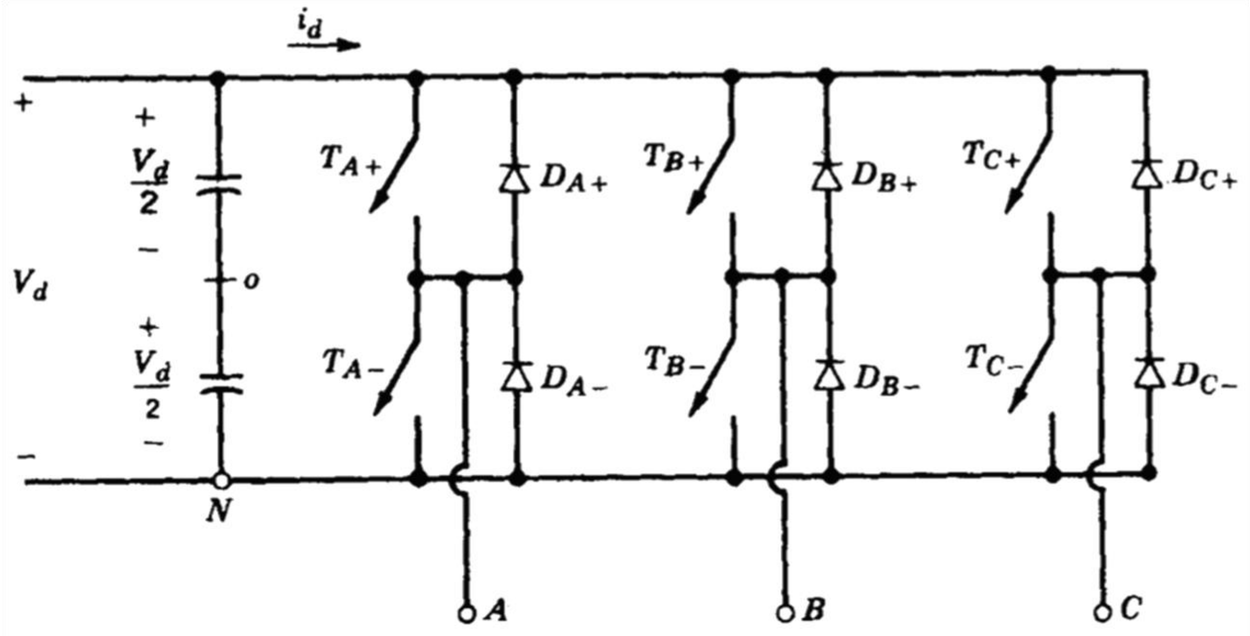
Figures used in the literature for 3-Phase Inverter

Operation

- The most frequently used three-phase inverter circuit consists of three legs, one for each phase.
- Each inverter leg is similar to the one used for describing the basic half bridge inverter.
- The output of each leg, for example v_{AN} (with respect to the negative dc bus), depends only on V_d, and the switch status.



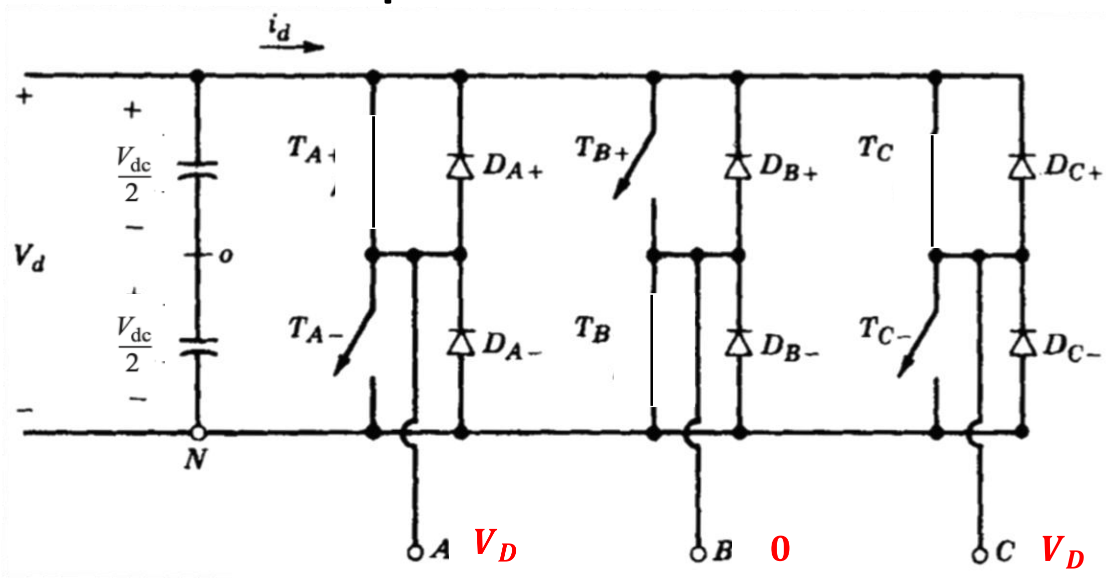
Operation – Square Wave



Note:

- The switches in each leg work in complementary way when square wave switching is applied i.e. if $\boldsymbol{T_{A+}}$ is closed than $\boldsymbol{T_{A-}}$ will be open and vice versa.
- Hence, information of one switch in one leg is given.

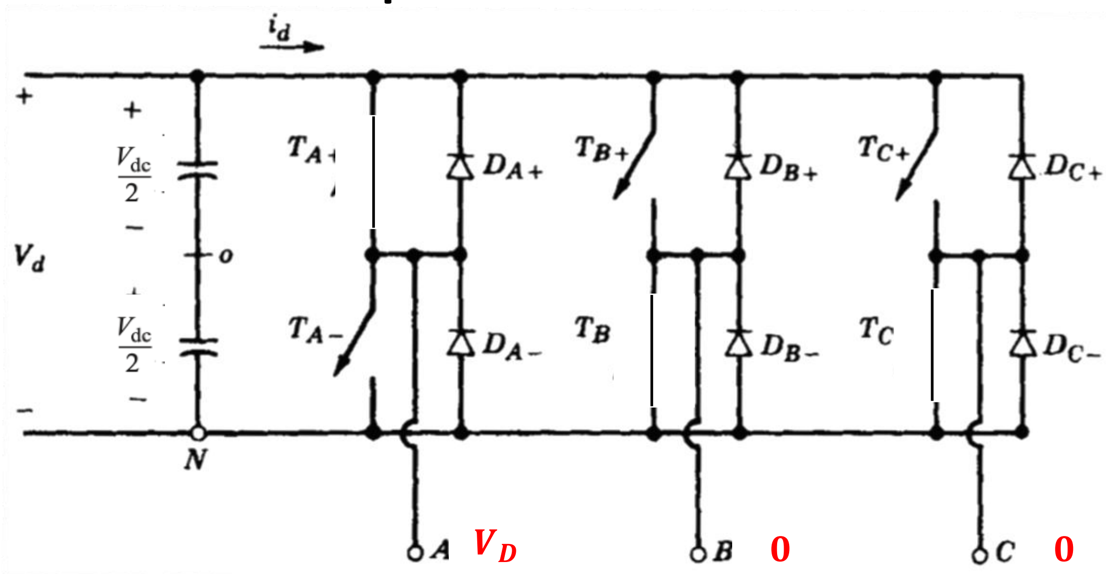
Operation – Square Wave



Seq. #	T_{A+}	T_{B+}	T_{C+}
1	Closed	Open	Closed
2	Closed	Open	Open
3	Closed	Closed	Open
4	Open	Closed	Open
5	Open	Closed	Closed
6	Open	Open	Closed

v_{AB}	v_{BC}	v_{CA}
V_D	$-V_D$	0

Operation – Square Wave



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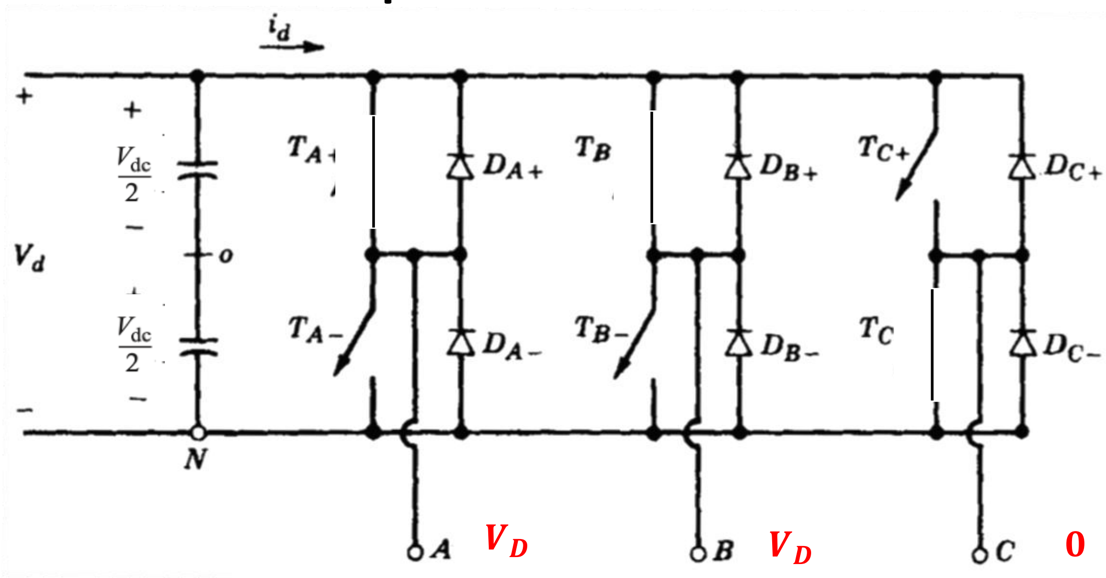
Operation – Square Wave



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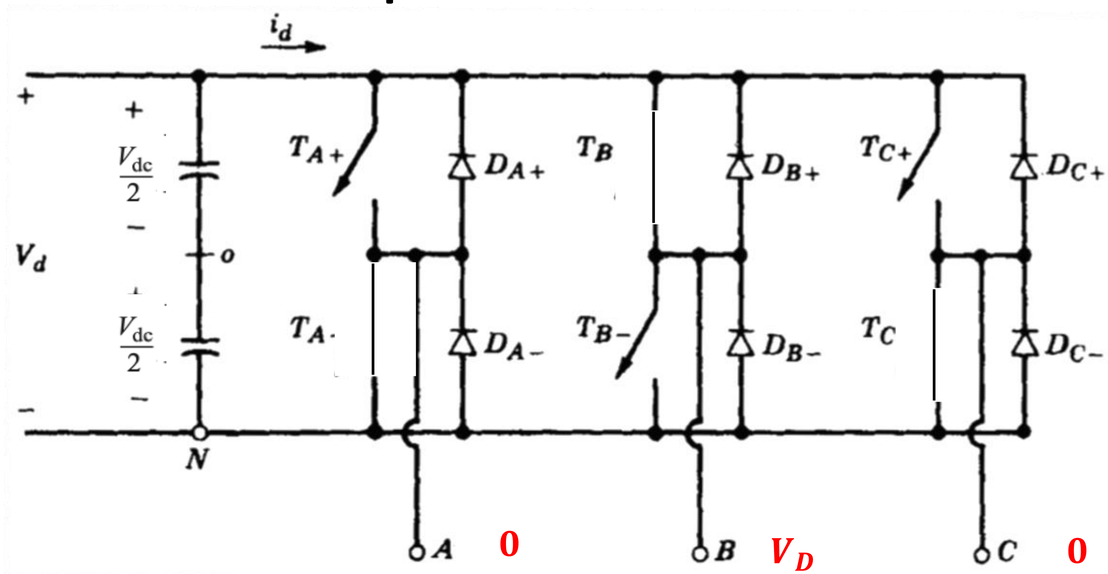
v_{AB}	v_{BC}	v_{CA}
V_D	$-V_D$	0
V_D	0	$-V_D$
0	V_D	$-V_D$

Operation – Square Wave



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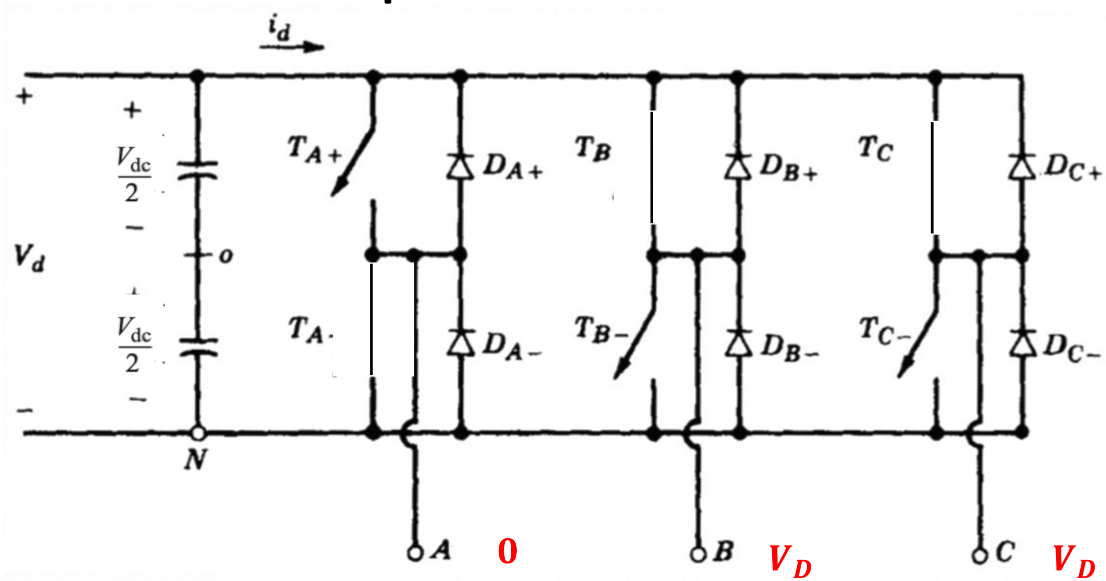
5	Open	Closed	Closed
6	Open	Open	Closed

v_{AB}	v_{BC}	v_{CA}
V_D	$-V_D$	0
V_D	0	$-V_D$
0	V_D	$-V_D$
$-V_{DC}$	V_D	0

Operation – Square Wave



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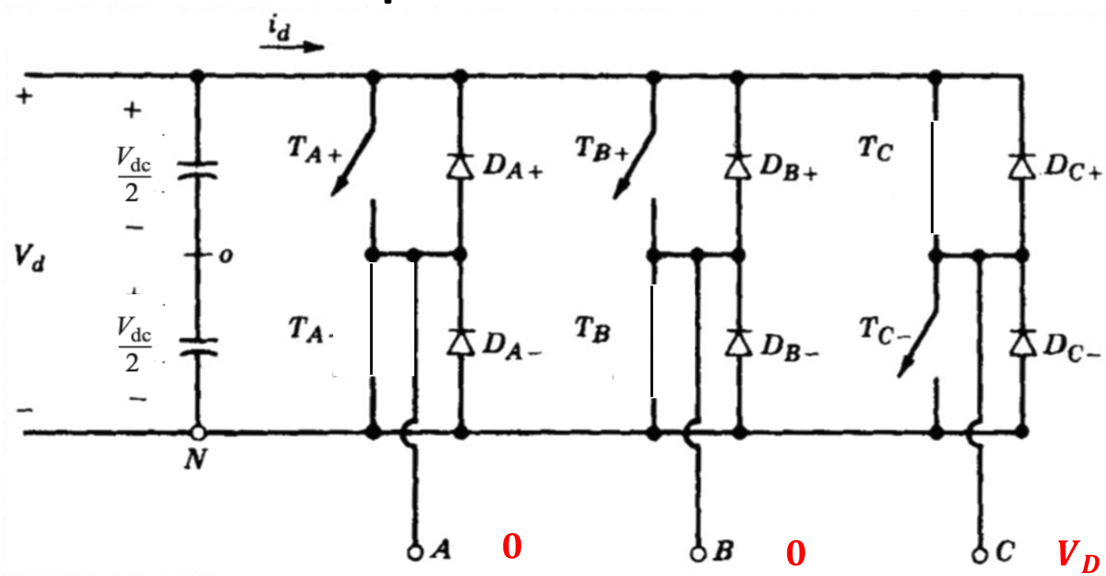
v_{AB}	v_{BC}	v_{CA}
V_D	$-V_D$	0
V_D	0	$-V_D$
0	V_D	$-V_D$
$-V_D$	V_D	0
$-V_D$	0	V_D

Operation – Square Wave



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v_{AB}	v_{BC}	v_{CA}
V_D	$-V_D$	0
V_D	0	$-V_D$
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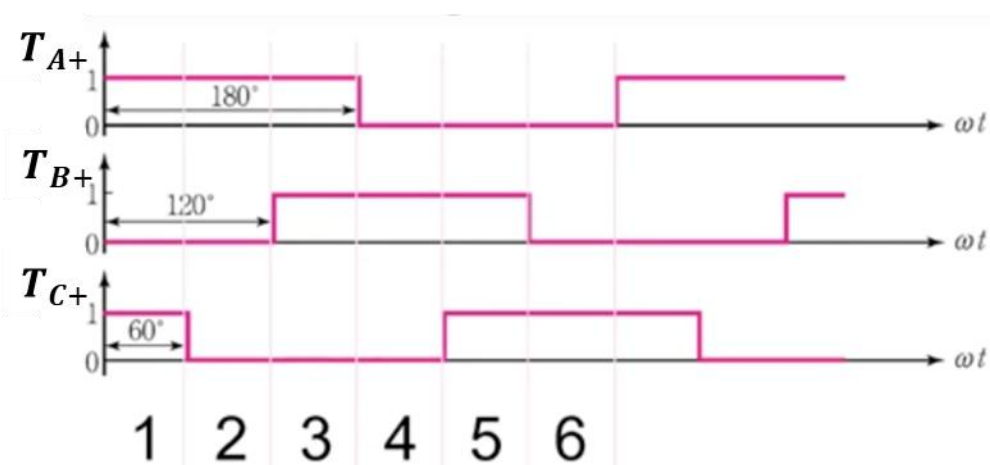
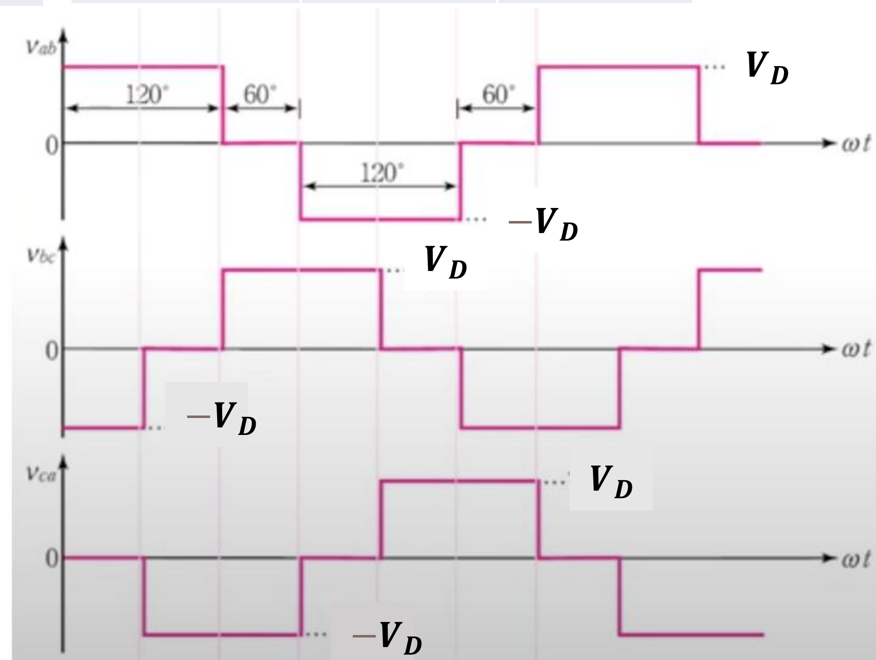
$-V_D$	V_D	0
$-V_D$	0	V_D
0	$-V_D$	V_D

Operation – Square Wave

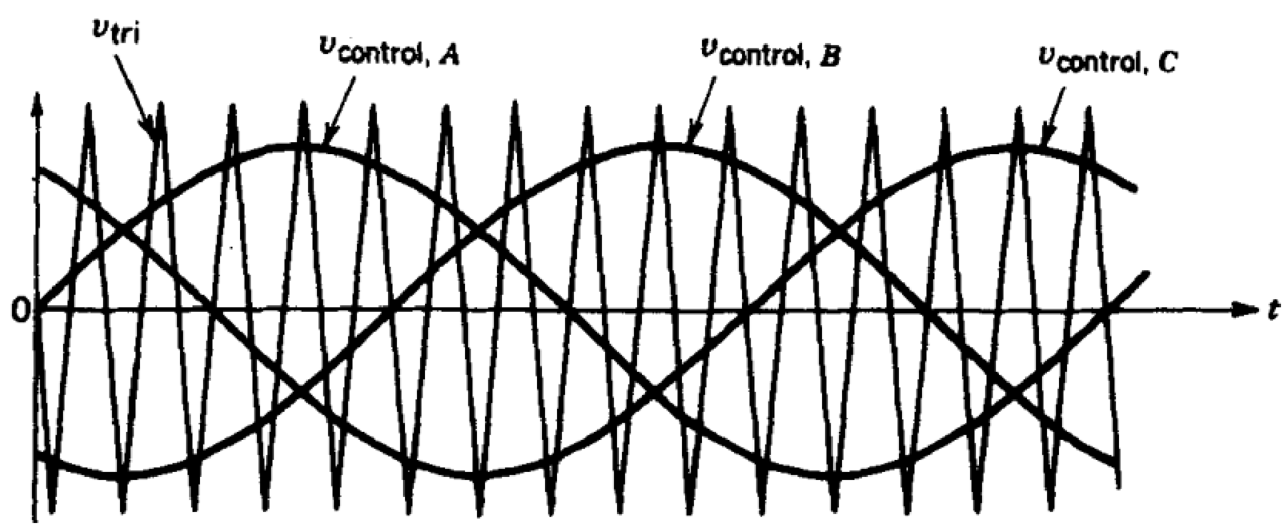
Seq. #	T_{A+}	T_{B+}	T_{C+}	v_{AB}	v_{BC}	v_{CA}
1	Closed	Open	Closed	V_D	$-V_D$	0
2	Closed	Open	Open	V_D	0	$-V_D$
3	Closed	Closed	Open	0	V_D	$-V_D$
4	Open	Closed	Open	$-V_D$	V_D	0
5	Open	Closed	Closed	$-V_D$	0	V_D
6	Open	Open	Closed	0	$-V_D$	V_D





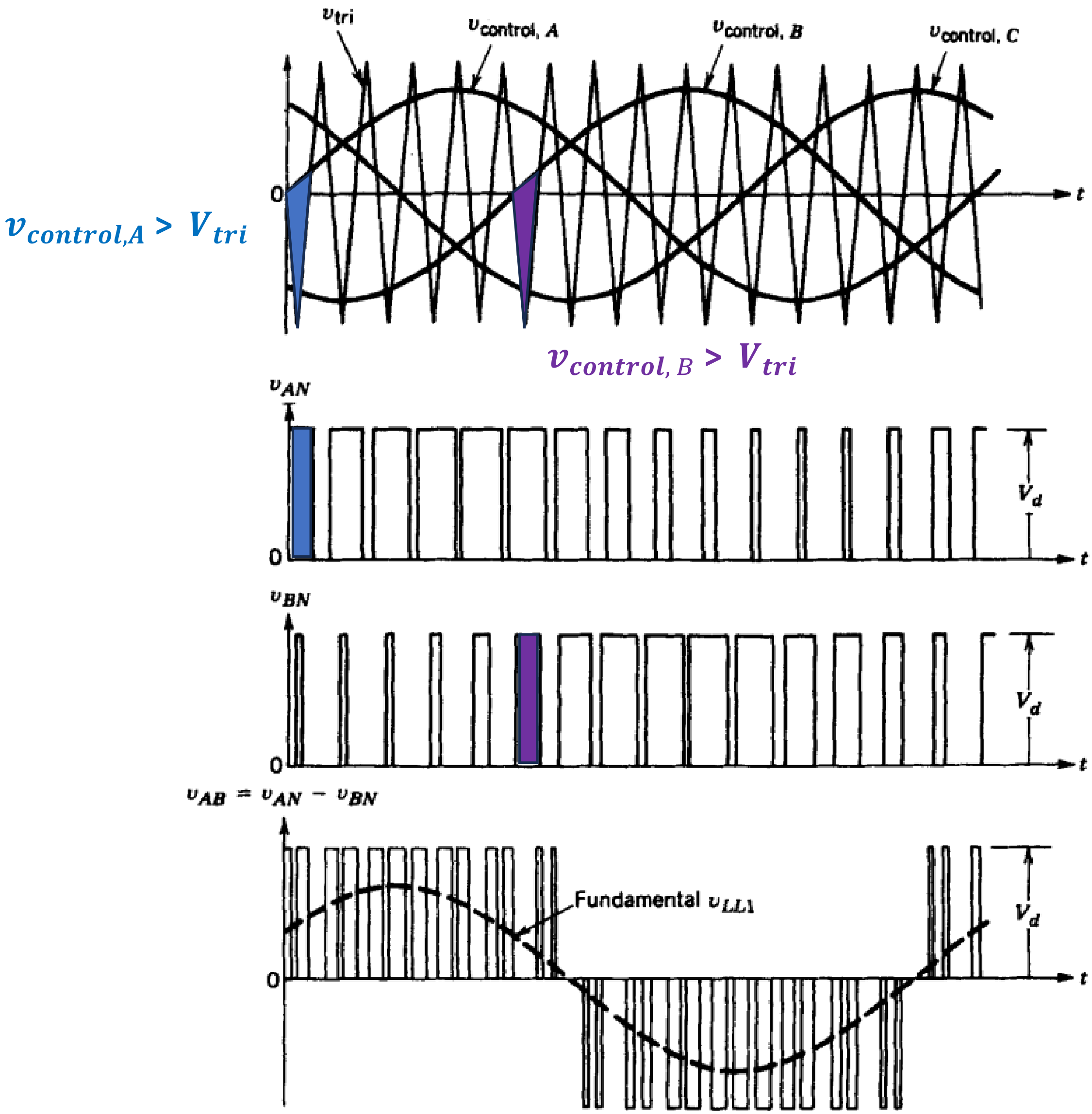
Operation

- Pulse Width Modulation (PWM) Switching
  - Is a type of switching technique.
  - Use two signal types called
    - v_{tri} (Triangular Signal)
      - Has frequency f_s (called switching frequency)
    - $v_{control,A}$, $v_{control,B}$ and $v_{control,C}$ (Control Signals)
      - Have frequency f_1 (desired fundamental frequency)



Operation

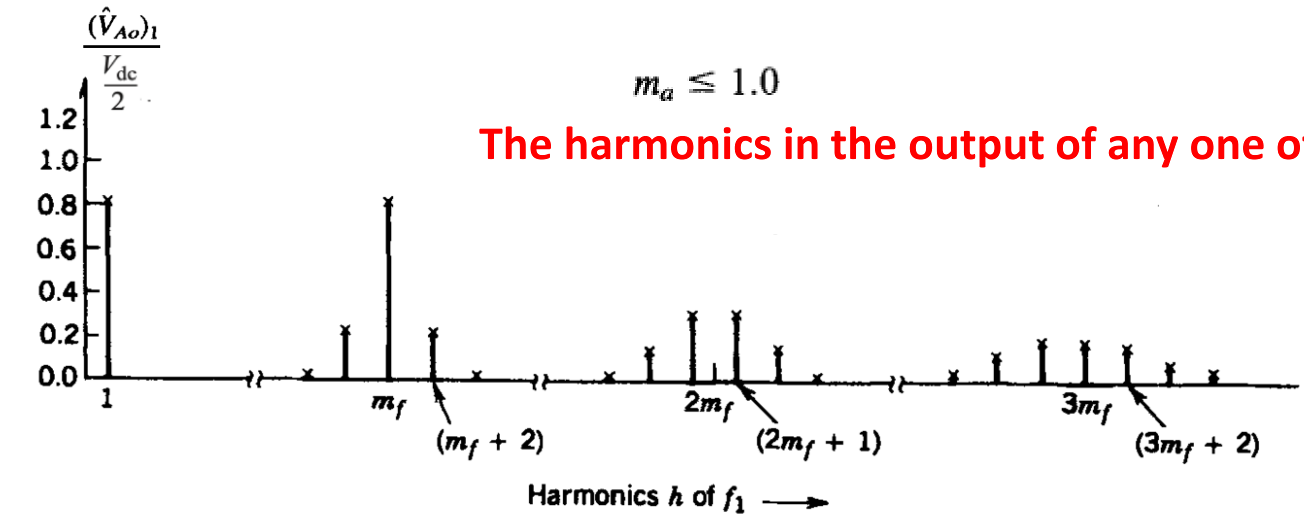
- Pulse Width Modulation (PWM) Switching



Harmonics

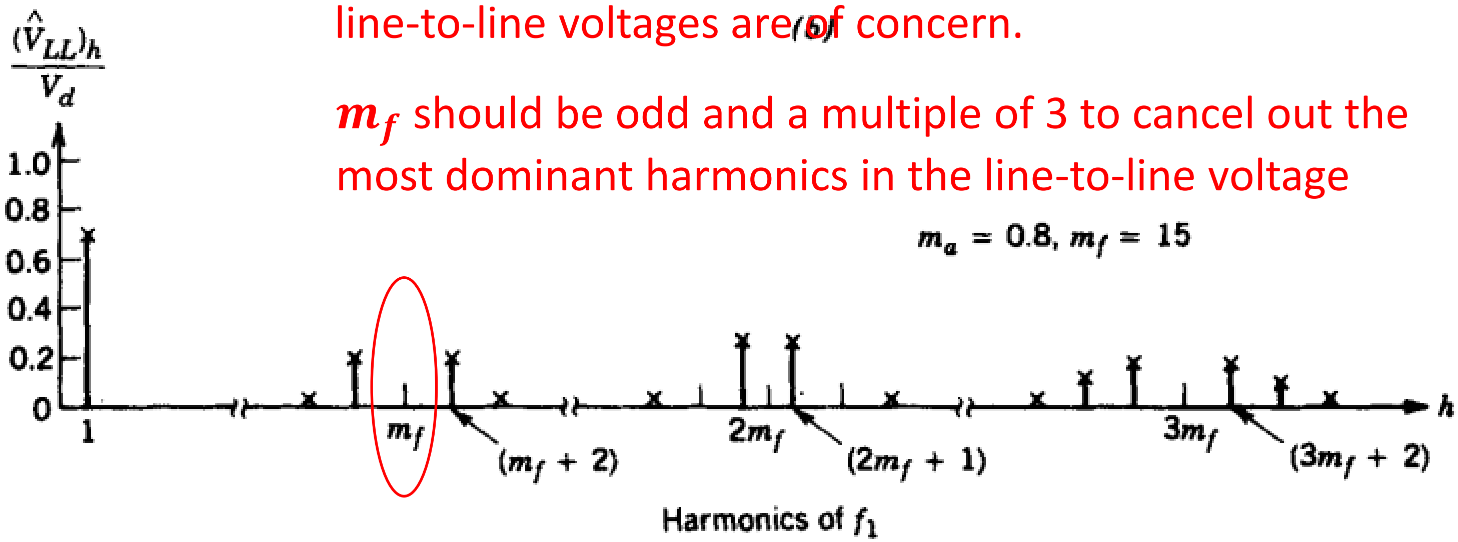
- Harmonics in PWM Switching

The harmonics in the output of any one of the legs



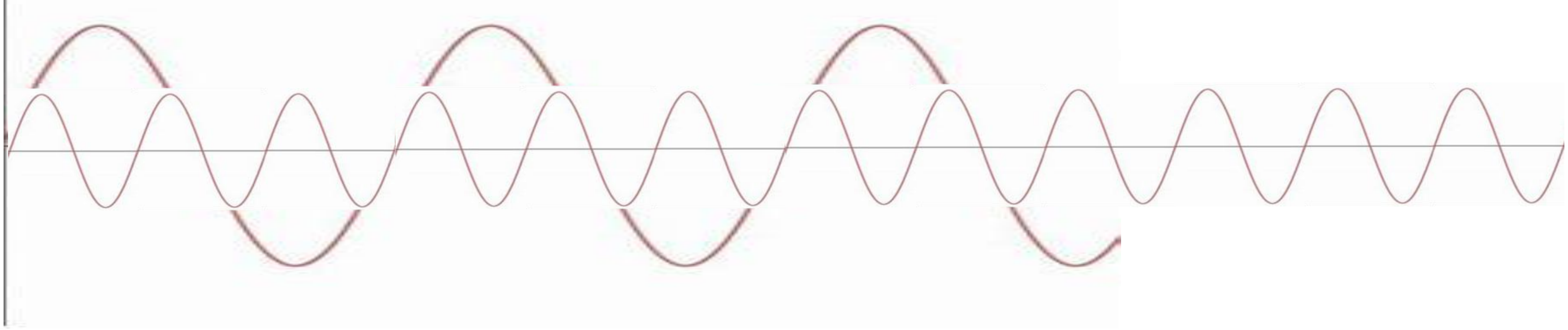
In the three-phase inverters, only the harmonics in the line-to-line voltages are of concern.

$\boldsymbol{m_f}$ should be odd and a multiple of 3 to cancel out the most dominant harmonics in the line-to-line voltage

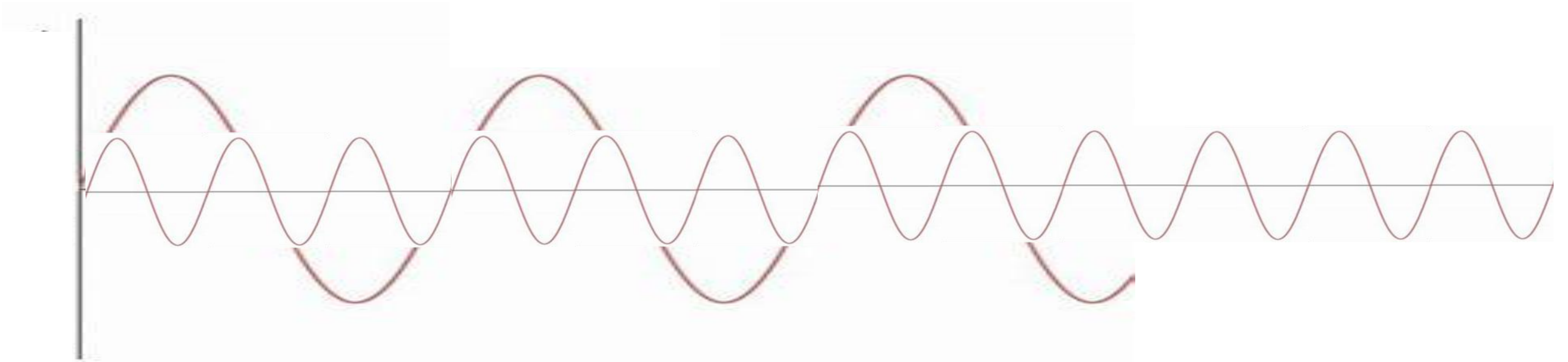


m_f = 3 which is odd and multiple of 3

Phase-A



Phase-B



Hence, this is how the third harmonic in v_{AB} will be eliminated from output

Practical Inverter

- The hardware used in half bridge inverter can be used for each leg of three phase inverter to control the switching.
- Similarly same techniques can be adopted to avoid the shoot through in both square wave and PWM switching mode.

Summary

- Single Phase Inverter Studied
  - Half Bridge
    - Square Wave Switching
    - PWM Switching
      - Operation, Output Voltage, Harmonics, Inverter made through real components and avoiding shoot through
  - Full Bridge
    - Square Wave Switching
    - PWM Switching
      - Bipolar
      - Unipolar
        - Operation, Output Voltage, Harmonics, Inverter made through real components and avoiding shoot through
- Three Phase Inverter
  - Square Wave Switching
  - PWM Switching
    - Operation, Output Voltage, Harmonics etc.

University of Glasgow

Power Electronics

Applications and Systems

Dr Atif Jafri
Atif.Jafri@glasgow.ac.uk

Applications - High level

- Topics covered so far
  - Power Supplies
  - Inverters

- Applications
  - Residential applications
  - Industrial applications
  - Electric Utility applications
  - Motor drive applications
  - Electric Vehicles

Residential Applications

- Residential homes and buildings are the endpoint of approximately 35% of the total electricity generated in the United States, which corresponds to approximately 8.5% of the total primary energy usage.

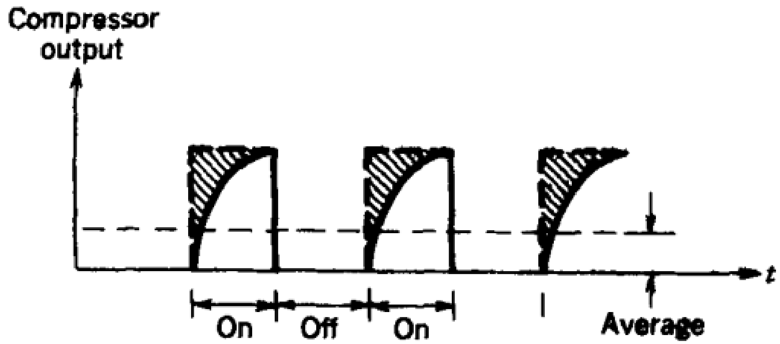
- The residential applications include space heating and air conditioning, refrigeration and freezing, water heating, lighting, cooking, television, clothes washer and dryer, and many other miscellaneous appliances.

SPACE HEATING AND AIR CONDITIONING

- Approximately 25-30% of the electric energy in an all-electric home is used for space heating and air conditioning.

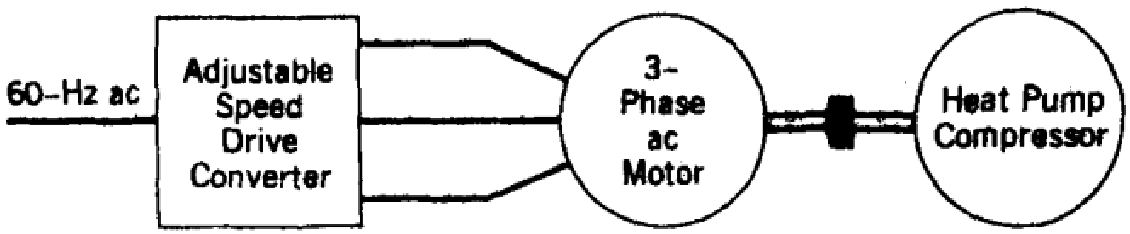
  - In a conventional heat pump, the compressor operates essentially at a constant speed when the motor is running.

  - The compressor output in this system is matched to the building heating or cooling load by cycling the compressor on or off



SPACE HEATING AND AIR CONDITIONING

- In a load-proportional capacity modulated heat pump, the speed of the compressor motor and hence the compressor output are adjusted to match the building heating or cooling load.

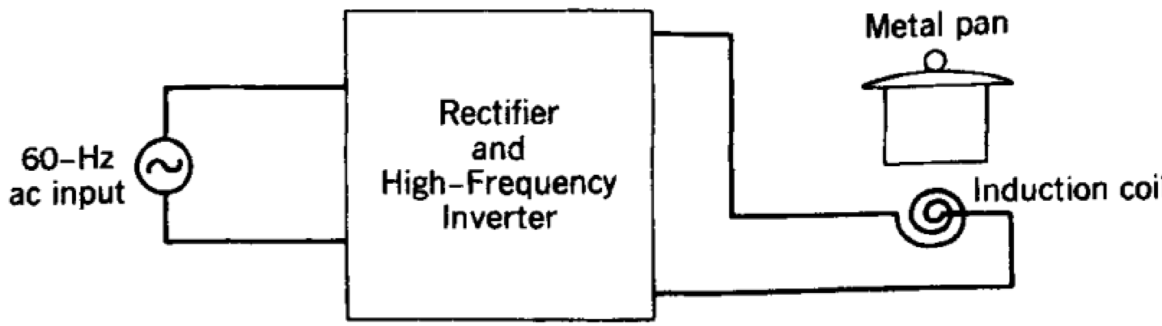


- Eliminating the on or off cycling of the compressor.

- The loss in the compressor output due to on-off cycling is eliminated in the load proportional capacity-modulated heat pump, where the speed of the compressor and, hence, the compressor output are adjusted to equal the building load.

INDUCTION COOKING

- In a standard electric or gas cooking range, a significant amount of heat escapes to the surroundings, thus resulting in poor thermal efficiency.



- The 60-Hz ac input is converted to a high-frequency ac in a range of 25-40 kHz, which is supplied to an induction coil.
- This induces circulating currents in the metal pan on top of the induction coil, thus directly heating the pan.

applications
welding.

Induction Heating

- In induction heating, the heat in the electrically conducting workpiece is produced by circulating currents caused by electromagnetic induction.

- Induction heating is clean, quick, and efficient.

- It allows a defined section of the workpiece to be heated accurately. The magnitude of the induced currents in workpiece decreases exponentially with the distance x from the surface, as given by the equation:

$$I(x) = I_0 e^{-x/\delta}$$

where I_0 is the current at the surface and δ is the penetration depth at which the current is reduced to I_0 times a factor $\frac{1}{e}$ (≈ 0.368).

- The penetration depth is inversely proportional to the square root of frequency f and proportional to the square root of the workpiece resistivity ρ

$$\delta = k\sqrt{\frac{\rho}{f}}$$

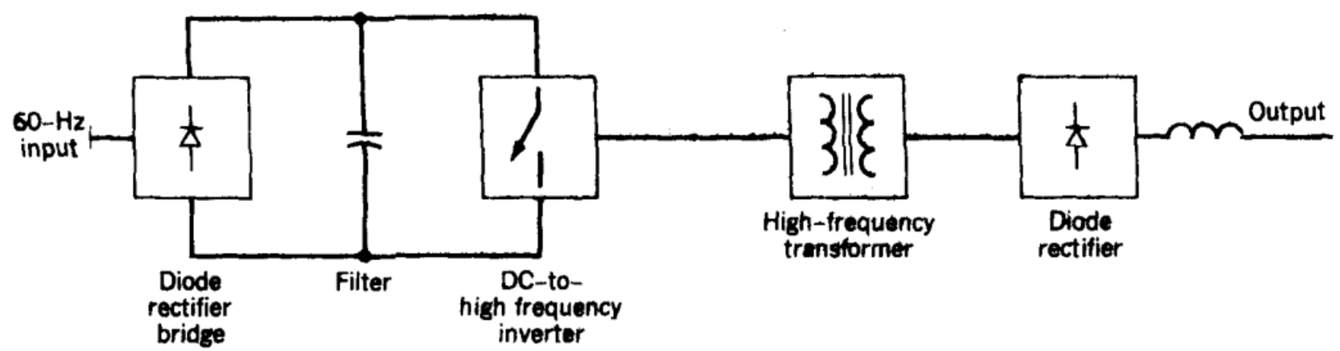
Induction Heating

$$I(x) = I_0 e^{-x/\delta} \qquad \delta = k\sqrt{\frac{\rho}{f}}$$

- The induction frequency is selected based on the application.
- A low frequency such as the utility frequency may be used for induction melting of large workpieces.
- High frequencies of up to a few hundred kilohertz are used for forging, soldering, hardening, and annealing.

Electric Welding

- In electric arc welding, the melting energy is provided by establishing an arc between two electrodes, one of which is the metallic workpiece being welded.
- The voltage-current characteristic of the welder depends on the type of welding process employed.
- Typical rated voltage and current are 50 V and 500 A dc, respectively.
- In all welding applications, the output needs to be electrically isolated from the utility input. This electrical isolation is provided by either a 60-Hz power transformer or a high.



Motor drive applications

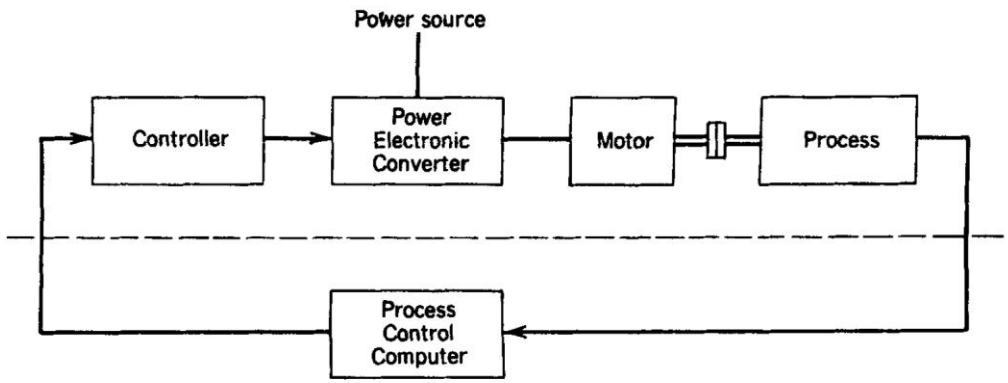


Figure 12-1 Control of motor drives.

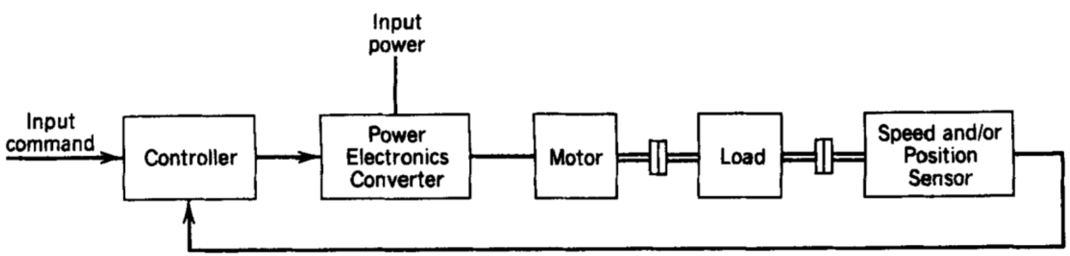
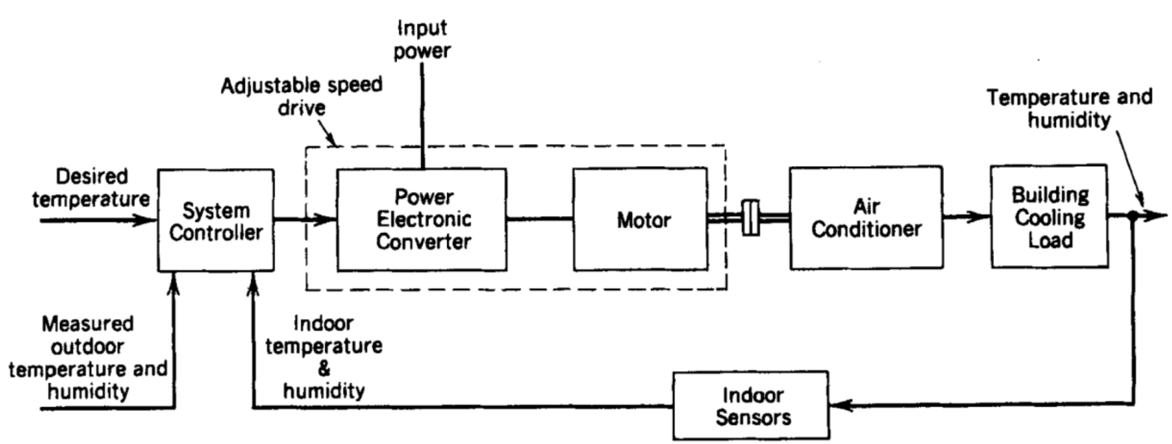
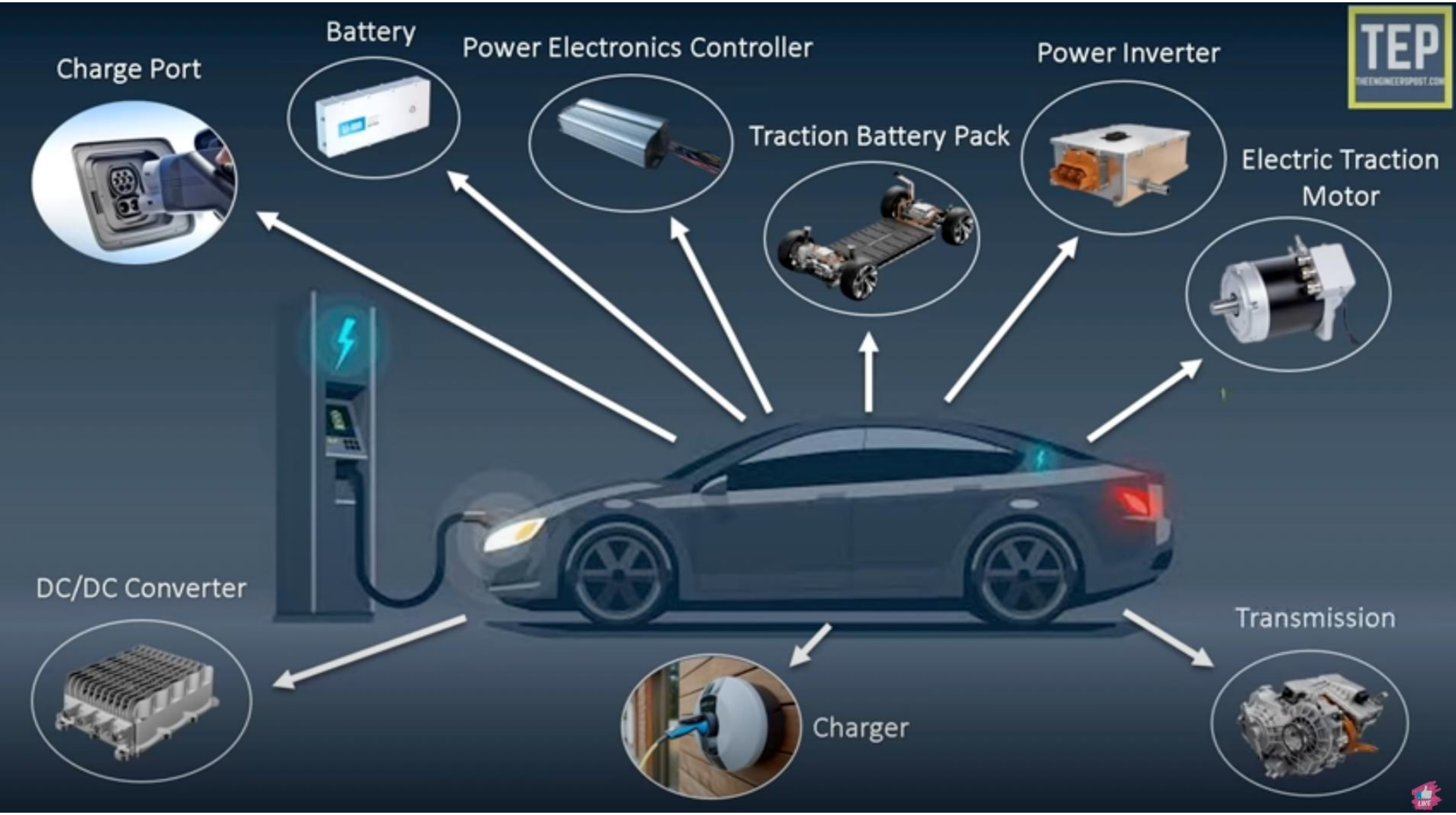
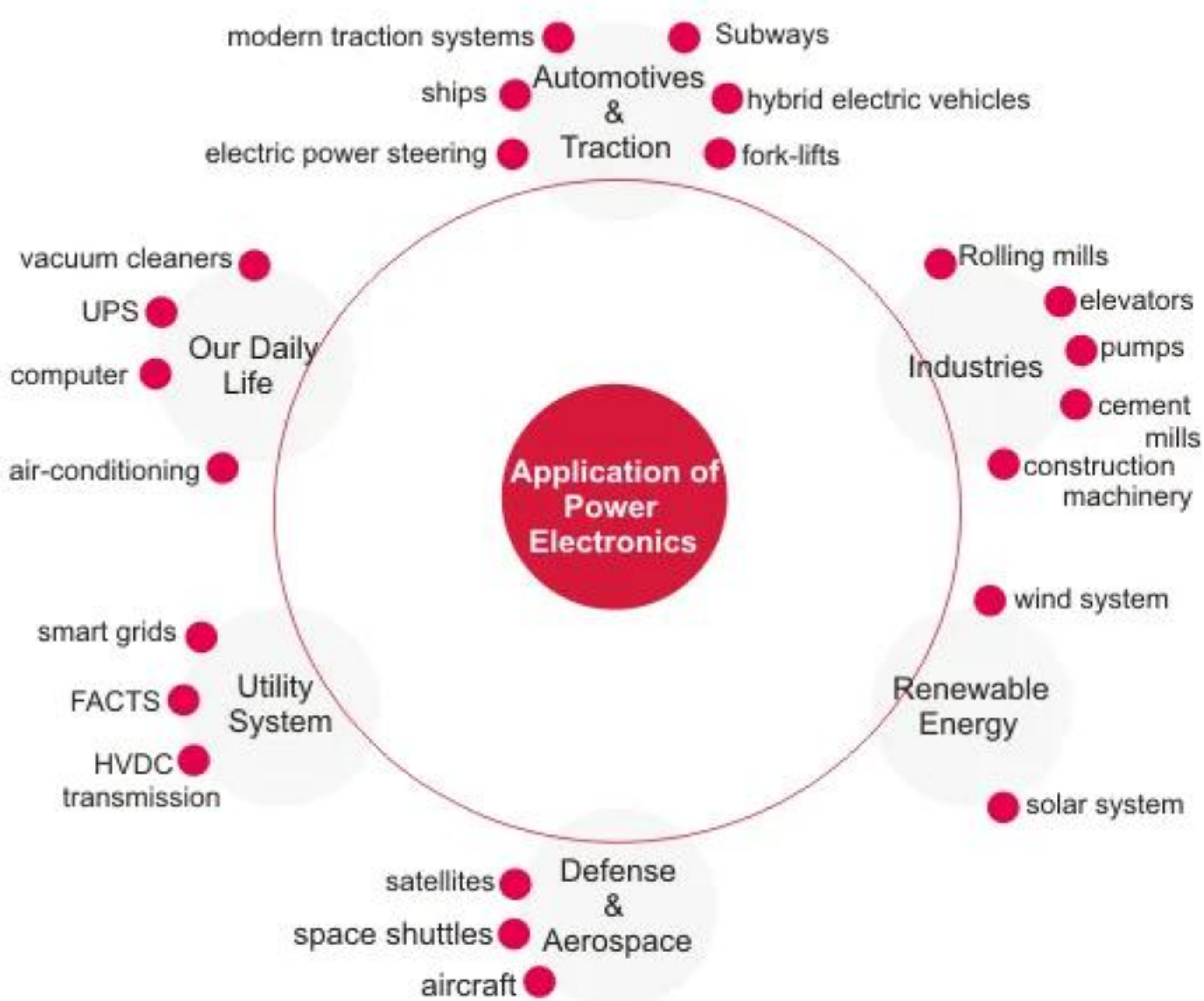


Figure 12-2 Servo drives.



Electric Vehicle





Application of Power Electronics
Automotives & Traction
modern traction systems
Subways
ships
hybrid electric vehicles
electric power steering
fork-lifts
Industries
Rolling mills
elevators
pumps
cement mills
construction machinery
Renewable Energy
wind system
solar system
Defense & Aerospace
satellites
space shuttles
aircraft
Utility System
smart grids
FACTS
HVDC transmission
Our Daily Life
vacuum cleaners
UPS
computer
air-conditioning

Additional links for self study

- Electric Vehicle – parts and functions
  - https://www.youtube.com/watch?v=tJfERzrG-D8
- Applications
  - https://www.youtube.com/watch?v=6KHJ5JA4BbY
- https://www.electrical4u.com/application-of-power-electronics/